

Manifold semi-supervised learning for aluminum electrolysis temperature identification based on regularized hierarchical extreme learning machine

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Yongxiang Lei¹ , Fang Liu², Hamid Reza Karimi¹ and Xiaofang Chen¹

Abstract

The aim of this article is to develop a soft approach for a real-time cell temperature prediction in the aluminum electrolysis reduction. Under the limited labeled data constraint, Laplacian semi-supervised learning methods, which can fully utilize the underlying structure of the data distribution and further extract information contained in all available data, has recently received extensive attention in the field of soft sensor modeling. Since the Laplacian underlying manifold is a constant, it remains a challenging task to improve the extrapolating ability for the case that only a few labeled samples are available. This study presents a soft modeling method based on a semi-supervised deep learning structure, which was developed from the hierarchical autoencoders with extreme learning machine. Furthermore, a Laplacian–Hessian semi-supervised extreme learning machine is built to learn all the geometric distribution information. The Laplacian–Hessian semi-supervised extreme learning machine method is applied to estimate the cell temperature in an aluminum reduction process. The experimental results demonstrate the performance and robustness of the proposed algorithm are superior to those of the existing state-of-the-art methods.

Keywords

Cell temperature identification, soft modeling, extreme learning machine, manifold semi-supervised learning, Hessian regularization

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Introduction

In modern industrial processes, some key variables represent important indicators related to product quality and production efficiency. However, it is quite difficult to measure most of these variables using physical instruments because of the strict and harsh limitations of measurement environments, high offline analyzer cost, and large delay. A soft sensor is a virtual sensing technique to estimate hard-to-measure key variables through certain easy-to-measure process variables.^{1–4} It provides a feasible and economical alternative to costly and impractical physical measurement devices. In addition, with the rapid development of distributed control systems, many data-driven soft sensor modeling techniques have been extensively studied and applied successfully to many domains with time savings and low cost, particularly in process monitoring.^{5,6} Many data-driven soft modeling methods, such as artificial neural networks (ANNs),^{7,8} principal component analysis

(PCA),⁹ and partial least squares (PLS),¹⁰ have extended the popularity for soft sensors. However, they can only utilize labeled data that contain both input and output samples, and extensive unlabeled data are discarded.

In the last decade, Huang et al. proposed the extreme learning machine (ELM) as a new training algorithm for the single hidden feedforward network (SLFN).¹¹ Compared with conventional approaches, ELM only solves an analytic matrix about the output weights between the input and hidden layers, whereas the input

¹Department of Mechanical Engineering, Politecnico di Milano, Milano, Italy

²School of Physics and Electronics, Central South University, Changsha, China

Corresponding author:

Yongxiang Lei, Department of Mechanical Engineering, Politecnico di Milano, Milano, 410083, Italy.
Email: lei.yongxiang@csu.edu.cn

weights and hidden neurons are randomly generated. By utilizing the squared loss of the prediction, the output weights training process can be translated into a regularized least (or ridge) problem, the solution for which is a closed form. Many recent studies have indicated that ELM offers comparable or even better performance in many classification and regression tasks.^{12–16} Lei et al.¹⁷ investigated the distributed convex optimization under persistent attacks with a hybrid system. ELM can also be extended to many specific applications with high computation efficiency and generalization ability.^{18–22} Some other efficient artificial models, such as fuzzy logic,⁹ genetic algorithms,¹⁰ and convolutional neural network (CNN),¹¹ have been widely used in the application domains.

In recent years, deep learning (DL) has become a hot topic for complex feature extraction in process modeling. A layer-by-layer unsupervised learning method with deep networks has been used in many soft modeling processes. DL methods extend single hidden layer networks and use multilayer architecture to achieve remarkable feature representation. It has been confirmed that DL architectures such as deep soft networks^{22–26} and long-short time memory (LSTM),²⁷ which are especially suited for processing soft sensors, have exhibited excellent performance. However, these DL methods are mostly trained by a backpropagation (BP) algorithm and all the parameters must be fine-tuned for the entire deep network, which is commonly a time-consuming process. Wong²⁸ considered the outlier in constructing a Roller Bearing Degradation Assessment method based on a deep MLP convolution neural network. Min et al.¹⁹ proposed a semi-supervised hierarchical ELM (SS-HELM) that significantly decreases computational cost as compared to the traditional DL methods in terms of quality variable prediction tasks. Recently, DL modeling in industrial processes have been introduced for soft sensor and monitoring tasks.^{29–32} In addition, Laplacian regularization has been incorporated as a regularization term to penalize the objective function.³³ However, this Laplacian-based method lacks extrapolating ability and may exhibit fast fading performance when only a few labeled data are available. Yuan et al. proposed a DL feature representation based on variable-wise weighted stacked autoencoder (SAE), which enhances the prediction performance as compared to traditional multilayer neural network (NN) and SAE. Lei et al.¹⁷ analyzed the performance difference of shallow neural network toward deep architectures, and they also built the latent variables model for regression prediction. Lei et al.³⁴ investigated the inner knowledge of the aluminum production and built a semi-supervised extreme learning method. They fused the deep complex features and density features in training image samples.²⁹ Although

several literatures have been reported, DL is in its initial stage for soft sensor domains, some problems are still unsolved and more studies are thus required.^{10–14} For example, based on Laplacian regularization, SS-ELM can determine the geometric structure of underlying manifolds even as it lacks extrapolating power and biases of the solution toward a constant.^{15–17}

In this study, a deep hybrid regularization semi-supervised architecture is proposed for industrial soft sensor modeling processes. We can use all labeled and unlabeled data to establish the entire modeling. First, the deep autoencoder (AE) structure is utilized to extract features included in all available process data using an unsupervised hierarchical learning method. Then, a semi-supervised ELM is further used in quality variable prediction with all available process data including both labeled and unlabeled data. In addition, Laplacian and Hessian regularization methods are integrated into semi-supervised soft modeling training to improve the extrapolating power and robustness. Fu et al.³¹ introduces a single-layer hybrid network for superheat degree classification in aluminum electrolysis production. In this article, a deep extended efficient autoencoders network for the cell temperature (CT) regression, is proposed. In particular, the following contributions have been made in this article.

- An efficient soft sensor modeling algorithm that can significantly improve computational efficiency is proposed. The developed model constructs a deep architecture that fully enhances the Laplacian and Hessian terms for improving the extrapolating ability and robustness.
- By adding unlabeled data, the accuracy of the proposed semi-supervised soft sensor model is superior to traditional supervised soft sensor models.
- Different from the study by Shen et al.,²¹ a deep AE is built for extracting complex features and learning in its hidden weights by the layer-to-layer low-computation training alternatives.

To evaluate the performance, the Laplacian–Hessian SS-HELM (LHSS-HELM) is applied to the industrial CT prediction in aluminum electrolysis.

The remaining sections of this article is organized as follows. The section “Preliminaries” specifies the foundation of challenges, AE and hierarchical AE (HAE) network, Laplacian regularization, and Hessian regularization. The section “LHSS-HELM Soft Modeling” introduces the proposed LHSS-HELM formation. Experiments on an industrial application in soft modeling are described in the section “Application of Cell Temperature Soft Modeling for an Aluminum Reduction Process.” Finally, Section “Conclusion” concludes the article.

Preliminaries

In this section, we briefly introduce ELM, the AE and HAE network, Laplacian regularization, which lay the foundations for soft sensor research.

Problem description

The CT is a vital indicator that can reflect the whole production situation of the aluminum electrolysis process. Keeping the CT in a suitable range can guarantee the high efficient production and extend the life span of the bath. However, in the practical production, the detection of CT is mainly based on the thermocouple, which is a device with expensive cost and time delay. Furthermore, the method of detection measures the local surface temperature that cannot reflect the whole field of the aluminum production. An analysis of the knowledge mechanism indicates that the inner links are existing in these processes variables, which means some data-driven model can be proposed.

AE and HAE network

An AE is a type of unsupervised one hidden layer neural network that can map the output layer to be equal to the input layer as accurately as possible. Figure 1 describes the structure of an AE. Assume that the input dimension of the AE is $\mathbf{x} = [x_1, x_2, \dots, x_{n_i}]^T$, where n_i is the dimension of the input variables. First, the AE projects from the input layer to the hidden layer by the mapping function as follows

$$\mathbf{H} = f(\mathbf{W}\mathbf{x} + \mathbf{b}) \quad (1)$$

where $\mathbf{H} = [h_{(1)}, h_{(2)}, \dots, h_{(n_h)}]^T$ is the hidden representation and n_h is the dimension of hidden neurons. $\mathbf{W}$ is a weight index matrix with the dimension of $n_i \times n_h$ and $\mathbf{b}$ is the bias vector. The nonlinear activation function of f can be chosen from the sigmoid, tanh, and rectified linear unit functions. Then, the hidden layer matrix $\mathbf{H}$ is mapped to the output layer by the function g

$$\hat{\mathbf{x}} = g(\hat{\mathbf{W}}\mathbf{H} + \hat{\mathbf{b}}) \quad (2)$$

where $\hat{\mathbf{W}} \in \mathbf{R}^{n_h \times n_o}$ is the weight matrix that connects the hidden layer to the output layer and $\hat{\mathbf{b}}$ is the n_i dimension bias vector. The whole hyperparameter must be fine-tuned as $\theta = \{\mathbf{W}, \hat{\mathbf{W}}, \mathbf{b}, \hat{\mathbf{b}}\}$. Then, the loss function can be defined by a square loss function as follows

$$L_{AE}(\theta) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad (3)$$

The parameters of the AE are iteratively updated by the gradient descent algorithm. HAEs represent a deep

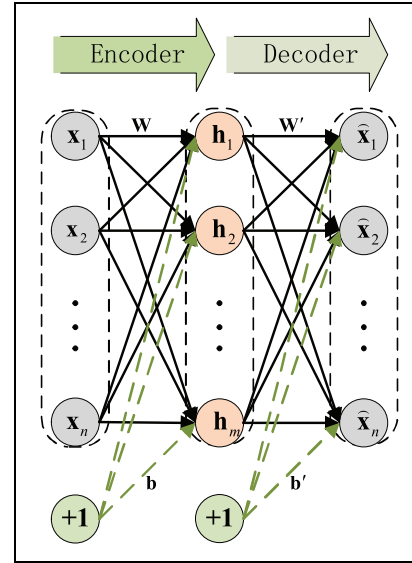


Figure 1. Single hidden structure of an AE.

network with multiple AEs connected layer by layer. Two stages are conducted for HAE training: layer-wise pre-training and supervised fine-tuning. In the first stage, the first AE maps the input data to the first hidden layer to obtain the first feature by minimizing the reconstruction error. Then, the output of the first hidden layer is further utilized as the input data for the second AE. All AEs are pre-trained layer by layer in this manner. In the second stage, an output layer is added to the topmost HAE to fine-tune the entire deep network. The pretrained weights serve as the initialization of the hidden layer weights, and then BP is utilized in the last fine-tuning process to obtain the superior weight. The loss function of the predicted error and the target variable is defined as follows

$$L_{HAE} = \sum_{i=1}^N \|y_i - \hat{y}_i\|^2 / 2N \quad (4)$$

where y_i and $\hat{y}_i$ are the labeled value and predicted output of the i th sample, respectively.

Laplacian regularization

The following two assumptions are imposed on a semi-supervised learning framework:

Assumption 1: Both labeled data $\mathbf{X}_l$ and unlabeled data $\mathbf{X}_u$ are collected from the same marginal distribution P_x .

Assumption 2: The conditional probabilities of $P(\mathbf{y}|\mathbf{x}_1)$ and $P(\mathbf{y}|\mathbf{x}_2)$ should be similar if the two points of $\mathbf{x}_1$ and $\mathbf{x}_2$ are close to each other.

The Assumption 2 is known as a smooth assumption and is widely used in machine learning. We get the

objective regularization function by using the smoothness assumption on the data as follows

$$L_m = \frac{1}{2} \sum_{i,j} u_{ij} \|P(\mathbf{y}|\mathbf{x}_i) - P(\mathbf{y}|\mathbf{x}_j)\|^2 \quad (5)$$

where u_{ij} measures the pair-wise similarity between the two samples of $\mathbf{x}_i$ and $\mathbf{x}_j$. The similarity weights are usually computed by using the Gaussian kernel function $\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma^2)$ or are fixed to 1.

Equation (5) can be further transformed into the following matrix form

$$\hat{L}_m = \text{Tr}(\hat{\mathbf{Y}}^T \mathbf{L} \hat{\mathbf{Y}}) \quad (6)$$

where $\text{Tr}(\cdot)$ denotes the trace of a matrix and $\hat{\mathbf{Y}}$ is the predicted value of output. $\mathbf{L} = \mathbf{D} - \mathbf{U}$ is the graph Laplacian and $\mathbf{D}$ is a diagonal matrix with diagonal elements

$$U_{ii} = \sum_{j=1}^{l+u} u_{ij}.$$

LHSS-HELM soft modeling

In this section, we first demonstrate the improved Hessian term, and then analyze the Laplacian and Hessian terms; finally we present the LHSS-HELM algorithm in detail.

Hessian regularization

The Hessian is a regularization term to learn the manifold energy. Define the smooth manifold $\mathbf{M} \subset \mathbb{R}^n$ and its local tangent space $\mathbf{T}_{\mathbf{x}_i}^*(\mathbf{M})$ for point $\mathbf{x}_i \in \mathbf{M}$, $\varphi(\mathbf{x}_i) = \mathbf{f}_i$ is a projection that can get the predictive value of $\mathbf{x}_i$. Then, the Hessian energy $E_H(\varphi)$ is expressed as

$$E_H(\varphi) = \int_{\mathbf{M}} \|\nabla_a \nabla_b \varphi\|^2_{\mathbf{T}_{\mathbf{x}_i}^* \mathbf{M} \otimes \mathbf{T}_{\mathbf{x}_i}^* \mathbf{M}} dV(\mathbf{x}) \quad (7)$$

where $\nabla_a \nabla_b \varphi$ is the second derivative of φ , and $dV(\mathbf{x})$ represents the natural volume element. The Hessian energy function in equation (7) is then transformed into a normal coordinate efficient on $\mathbf{M}$. In a normal covariant x^r centered at $\mathbf{x}_i$, the following equation is obtained

$$\|\nabla_a \nabla_b \varphi\|^2_{\mathbf{T}_{\mathbf{x}_i}^* \mathbf{M} \otimes \mathbf{T}_{\mathbf{x}_i}^* \mathbf{M}} = \sum_{r,s=1}^l \left(\frac{\partial^2 \varphi}{\partial x^r \partial x^s} \right)^2 \quad (8)$$

As shown in the study by Kolar et al.,¹¹ the final Hessian energy is approximated as

$$\hat{E}_{H(\varphi)} = \sum_{i=1}^n \sum_{\alpha \in N_k(\mathbf{x}_i)} \sum_{\beta \in N_k(\mathbf{x}_i)} \mathbf{f}_\alpha \mathbf{f}_\beta H^{(i)}_{\alpha\beta} = \mathbf{F}^T \mathbf{H} \mathbf{F} \quad (9)$$

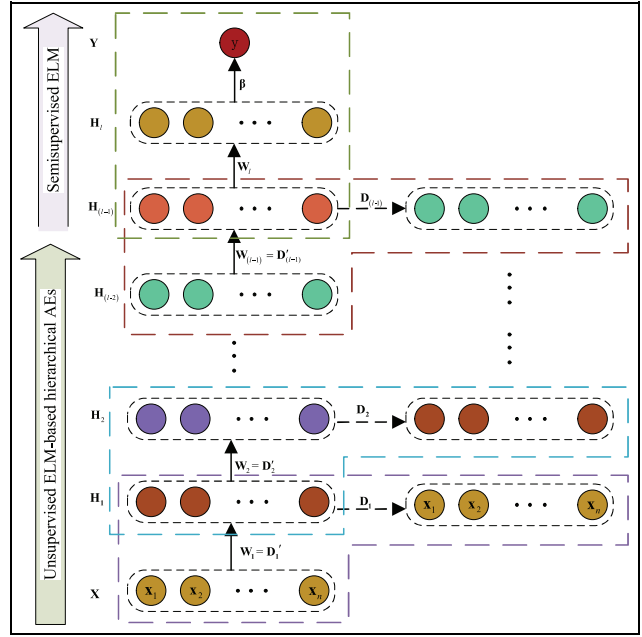


Figure 2. LHSS-HELM network structure. The first $l-1$ hidden layers are constructed by the unsupervised HAEs. The semi-supervised ELM is in the top level.

where $\mathbf{H} = \sum_{i=1}^n H^{(i)}$ and n is the amount of training data.

Analysis of Laplacian and Hessian terms

Although Laplacian regularization has achieved good results, the solution of the Laplacian method is biased toward a constant and lacks extrapolating power. Regularization based on estimation of the Hessian favors mapping functions whose values vary linearly along the geodesic distance, and it preserves the local manifold structure better than the Laplacian. However, Hessian regularization is not robust to many tasks, particularly when the function is oscillating, which decreases the quality of the local fit. Therefore, the performance of Hessian-based models is not always accurate with real-world tasks. Recently, Liu et al. introduced Laplacian–Hessian terms for semi-supervised learning problems, which provides some insight into building the LHSS-HELM. Inspired by the above analysis, we extend the Laplacian–Hessian regularization to deep AEs and propose the LHSS-HELM soft sensor model.

LHSS-HELM formation

The structure of the proposed LHSS-HELM is presented in Figure 2. First, one can train the HAEs to obtain multilayer weights $\mathbf{W}_i (i = 1, \dots, h)$ in an unsupervised learning method with all the available data. The parameters of the AE are obtained as an ELM method

as given in equation (2). All the hidden weights parameters are obtained layer by layer. To extract deep and complex features, HAEs are constructed and the parameters are trained layer by layer. Compared to the BP algorithm, the network parameters are obtained in a closed solution rather than an iterative way, then the computational complexity is greatly improved. Furthermore, for top-layer HAE-based ELM training, Laplacian and Hessian regularization are integrated to take advantage of the manifold structure of the data space to improve the performance of the traditional ELM. We rewrite equation (5) as follows

$$\begin{aligned} \Gamma_{\text{LHSS-ELM}} &= \min_{\beta \in \mathbb{R}^{n_h \times n_o}} \frac{1}{2} \|\beta\|^2 + \frac{C}{2} \|\mathbf{e}_i\|^2 \\ &\quad + \frac{\lambda_1}{2} \text{Tr}(\hat{\mathbf{Y}}^T \mathbf{L} \hat{\mathbf{Y}}) + \frac{\lambda_2}{2} \text{Tr}(\hat{\mathbf{Y}}^T \mathbf{H} \hat{\mathbf{Y}}) \\ \text{s.t.} \quad \mathbf{e}_i &= \mathbf{y}_i^T - \mathbf{J}(\mathbf{x}_i) \beta, i = 1, \dots, l \\ \hat{\mathbf{Y}}_i &= \mathbf{J}(\mathbf{x}_i) \beta, i = 1, \dots, l + u \end{aligned} \quad (10)$$

where $\mathbf{L} \in \mathbb{R}^{(l+u) \times (l+u)}$ and $\mathbf{H} \in \mathbb{R}^{(l+u) \times (l+u)}$ are Laplacian and Hessian matrices derived from labeled and unlabeled data, respectively. $\mathbf{Y}$ is the output matrix and $\beta \in \mathbb{R}^{n_h \times n_o}$ represents the output weights, whereas λ_1 and λ_2 are trade-off parameters. In our framework, we associate different coefficients C_i on the prediction errors with examples from skewed classes. As given in the study by Sussner and Campiotti,²² we alleviate poor generalization by using the penalty coefficient of $C_i = C_0/N_{t_i}$, where C_0 is a user-defined parameter as in conventional ELM and N_{t_i} is the number of training patterns in class t_i . Therefore, the samples from the dominant classes will not be overfitted while the patterns from the less-dominant class examples will not be

neglected simultaneously. By substituting the constraints into the objective function, we rewrite equation (10) in the following matrix form

$$\begin{aligned} \Gamma_{\text{LHSS-ELM}} &= \min_{\beta \in \mathbb{R}^{n_h \times n_o}} \frac{1}{2} \|\beta\|^2 + \frac{C}{2} \|\mathbf{Y} - \mathbf{J}\beta\|^2 \\ &\quad + \frac{\lambda_1}{2} \text{Tr}(\beta^T \mathbf{J}^T \mathbf{L} \mathbf{J} \beta) \\ &\quad + \frac{\lambda_2}{2} \text{Tr}(\beta^T \mathbf{J}^T \mathbf{H} \mathbf{J} \beta) \end{aligned} \quad (11)$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{(l+u) \times n_o}$ is an augmented training target whose first l rows are equal to $\mathbf{Y}_l$ and the remaining rows are set to 0. $\mathbf{C}$ is a diagonal matrix of $(l+u) \times (l+u)$, with its first l diagonal elements $[\mathbf{C}]_{ii} = C_i$ and the remaining elements equal to 0.

By setting the gradient of equation (11) to 0, we can calculate the output weights of LHSS-ELM as

$$\beta^* = (\mathbf{I}_{n_h} + \mathbf{J}^T \mathbf{C} \mathbf{J} + \lambda_1 \mathbf{J}^T \mathbf{L} \mathbf{J} + \lambda_2 \mathbf{J}^T \mathbf{H} \mathbf{J})^{-1} \mathbf{J}^T \mathbf{C} \mathbf{Y} \quad (12)$$

where $\mathbf{I}_{n_h}$ is an identity matrix of dimension $n_h \times n_h$. When the number of labeled data is less than the number of hidden neurons, the alternative solution of LHSS-ELM is

$$\beta^* = \mathbf{J}^T (\mathbf{I}_{l+u} + \mathbf{C} \mathbf{J} \mathbf{J}^T + \lambda_1 \mathbf{L} \mathbf{J} \mathbf{J}^T + \lambda_2 \mathbf{H} \mathbf{J} \mathbf{J}^T)^{-1} \mathbf{C} \mathbf{Y} \quad (13)$$

where the $\mathbf{I}_{(l+u)}$ is an identity matrix with $(l+u) \times (l+u)$ dimensions. The detailed algorithm procedure is given in Algorithm 1.

Algorithm 1 LHSS-HELM algorithm

Require: Labeled data $\{\mathbf{X}_l, \mathbf{Y}_l\} = \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^l$;

Unlabeled data $\mathbf{X}_u = \{\mathbf{x}_i\}_{i=1}^u$;

Penalty coefficient C_0 and trade-off parameter sequence λ ;

Ensure: The modeling mapping function $\hat{\mathbf{Y}} = f(\mathbf{x})$; The output prediction of testing patterns $\hat{\mathbf{Y}}$.

Start:

Step 1: Select the variables from the technician workers' experience

Step 2: Pretreat the data (normalization, standardization);

Step 3: Split data into training, validation, and testing data;

Step 4: Determine the number of HAEs and of neurons in each layer;

Step 5: Randomly initialize input weights and train HAEs with all the available input data, and obtain weight $\mathbf{W}_i$ for each layer;

Step 6: Construct the Laplacian term $\mathbf{L}$ and Hessian operator $\mathbf{H}$ from both labeled and unlabeled $\mathbf{X}$;

Step 7: Initiate an ELM framework of n_h hidden neurons with random weights;

Step 8: Calculate the output of the hidden neurons of ELM $\mathbf{J} \in \mathbb{R}^{(l+u) \times n_h}$;

Step 9: **If** $n_h \leq N$, compute the output weights β using equation (12);

Else compute the output weights β using equation (13);

Step 10: Test whether the recent model is the best model:

(a) **yes**, compute the modeling mapping function $\hat{\mathbf{Y}} = f(\mathbf{x}, \mathbf{W}_i (i = 1, \dots, h), \beta)$;

(b) **no**, update λ and return to Step 8;

Step 11: Predict the quality variable $\hat{\mathbf{Y}}$ for testing patterns.

end.

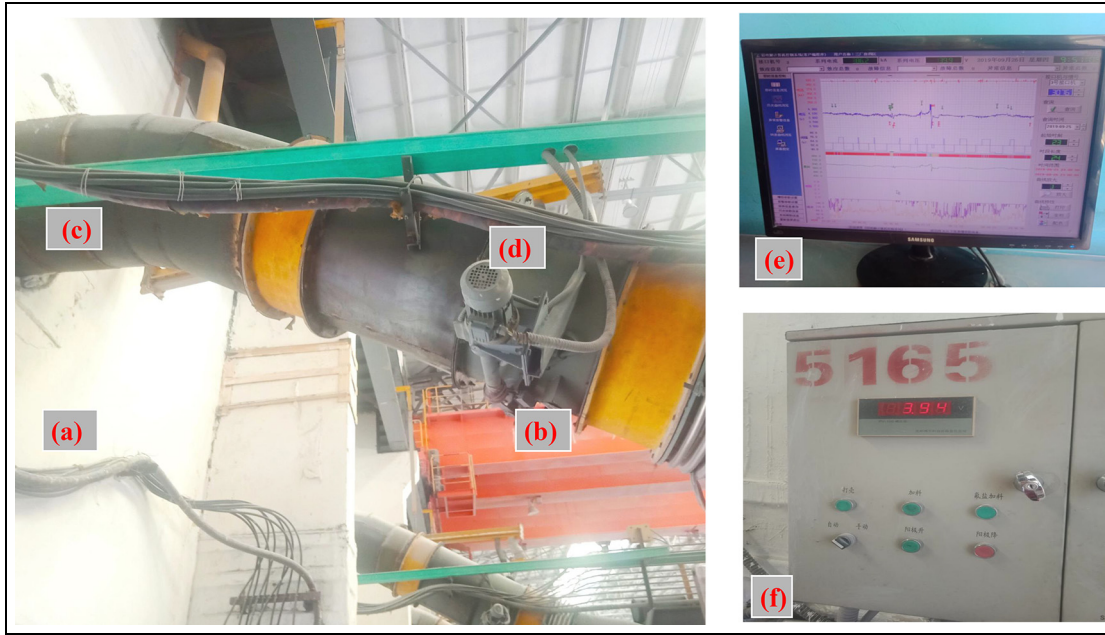
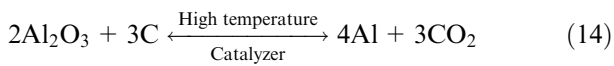


Figure 3. Schematic of an industrial aluminum reduction processes: (a) sensor signals turbines, (b) crane, (c) alumina powder channel, (d) motor, (e) PC, and (f) SY-CKJ600A control cabinet.

Application of CT soft modeling for an aluminum reduction process

In this section, the proposed regularization-integrated LHSS-HELM is applied to CT estimation in an industrial aluminum process. The aluminum reduction cell is the main unit to produce aluminum through electrolysis. The main reaction process in an aluminum bath is described as follows



The basic structure of an industrial aluminum reduction cell is shown in Figure 3. One of the main indices is the aluminum CT. The CT is an influential index derived from aluminum electrolysis production processes. Maintaining CT in a suitable range can sustain the physical field of the electrolytic cell and enhance the current efficiency and lifespan of the electrolytic cell. However, in the real process, the CT is measured by a technician using artificial physical devices that incur a high cost and are laborious. In addition, systematic errors caused by physical measurement devices and reading errors caused by field technicians are inevitable in CT measuring process. Real-time measurements and regression of CT remains a problem without a solution.

The root mean square error $RMSE$, training time (T), and variance (Var) index were used to evaluate the performance, computation cost, and robustness of the

LHSS-HELM soft processes modeling, respectively. $RMSE$ is defined as

$$RMSE = \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2 / (N - 1)} \quad (16)$$

where y_i and $\hat{y}_i$ are the labeled and predicted output value of testing examples, respectively, and N is the number of testing samples. A low $RMSE$ value indicates a superior performance.

Another variance index is defined as

$$Var = \sum_{i=1}^N (\hat{y}_i - \bar{y}_i) / N \quad (17)$$

where $\bar{y}_i$ is the average value of the testing sample output value. A small variance value indicates superior robustness of the algorithm.

To demonstrate the effectiveness of the proposed LHSS-ELM for CT estimation in an aluminum electrolysis cell, 13 process variables were selected based on the expertise and experience of technicians and operators. The average cell voltage and additional amount of AlF_3 in the reduction cell are related to the operating conditions in this process and would contribute most to the modeling performance. The relevant input variables of CT as selected by operators are described in Table 1.

Twelve input process variables are selected for training the proposed LHSS-ELM and both the input variables were normalized at $(-1,1)$, and the output CT was normalized at $(0,1)$. Also, 1200 samples are

Table 1. Process variables selected based on operator experience.

No.	Tags	Descriptions
1	u_1	Working current
2	u_2	Cell resistance
3	u_3	Set datum time of Al_2O_3
4	u_4	Actual feeding time of Al_2O_3
5	u_5	Content of Iron
6	u_6	Content of Silica
7	u_7	Aluminum-tapping level
8	u_8	Electrolyte level
9	u_9	Content of precipitated aluminum
10	u_{10}	Molecular ratio
11	u_{11}	Average cell voltage
12	u_{12}	Addition amount of AlF_3

collected from databases, 500 of which were used for training models, 300 of which are used to acquire the best model hyperparameters, and the remaining 400 are testing patterns. The parameters λ_1 and λ_2 are set based on their performances on the validation set. Based on the report by Li et al.,¹³ the penalty coefficient C is not sensitive to the manifold region so it was determined in the interval $[1, 10^-5]$ and $[5, 30]$ for HELM. To alleviate overfitting and improve the generalization ability, the penalty is adapted by the form $C_i = C_0/N_{t_i}$. The trade-off parameters λ_1 and λ_2 are tuned from the sequences set $\{10^{-8}, 10^{-7}, \dots, 10^{-2}\}$ and $\{10^3, 10^2, \dots, 10^{-8}\}$, respectively. Both λ_1 and C are determined by trial and error method in the empirical candidate set. The input neurons of ELM is set close to the input neurons. For the sparse representation of input variables, the hidden neurons were set to 1000 for ELM and 2000 for HELM. For HELM, the number of hidden neurons of SAE is less than the input neurons, which we set to 10. The hidden layer was set to 3 according to validation set results.²¹ All experimental algorithms are implemented utilizing MATLAB2016a with a 3.6 GHz i5 CPU and 64 GB RAM. Details of the parameter set are provided in Table 2.

A performance comparison using different unlabeled samples was conducted on the LHSS-HELM model to demonstrate that the extra scale unlabeled samples would greatly affect the prediction performance and robustness. We gradually increased the unlabeled data from 0 to 1000 to train the ELM, SS-ELM, HELM, and LHSS-HELM models. The 400 samples from the testing dataset were used to evaluate the prediction results. ELM and HELM employed 500 labeled samples for the supervised learning, and SS-ELM and LHSS-HELM are conducted under the same conditions for semi-supervised learning. Detailed CT prediction results are provided in Table 3, where N_u denotes the number of unlabeled patterns. We can observe that the accuracy rate was still enhanced with the increasing number of unlabeled samples and variance decreased simultaneously. This means that both the performance

Table 2. Parameters set for ELM, SS-ELM, HELM, and LHSS-HELM.

Methods	L_{sae}	C	λ_1	λ_2	N_{elm}	N_{sae}
ELM	—	5	0.1	0.2	100	—
SS-ELM	—	5	0.1	0.2	100	—
SS-HELM	3	25	0.1	0.2	200	10
LHSS-HELM	3	25	0.1	0.2	200	10

ELM: extreme learning machine; LHSS-HELM: Laplacian–Hessian semi-supervised hierarchical ELM.

Table 3. Comparison results with different methods. The bold font in Table 3 describes a superior experimental result.

Methods	Testing RMSE	Training time (s)	Variance
SS-PPCR	0.0069	16.91	0.0078
Co-PLS	0.0062	15.24	0.0071
DBN	0.0035	14.77	0.0041
LHRSS-ELM	0.0055	14.92	0.0056
SDAE-NN	0.0031	23.54	0.0034
LHSS-HELM	0.0029	15.42	0.0032

SS-PPCR: semi-supervised probabilistic principal component regression; PLS: partial least squares; DBN: deep belief network; LHRSS-ELM: Laplacian–Hessian regularized semi-supervised extreme learning machine; SDAE-NN: denoising autoencoder with a neural network; LHSS-HELM: Laplacian–Hessian semi-supervised hierarchical extreme learning machine.

and robustness are improved with the additional unlabeled training data. Some similar results could also be observed in the comparison of the basic ELM, HELM, and SS-ELM methods when only labeled training data are used, whereas LHSS-ELM degenerated to a universal ELM. This indicates that the proposed algorithm with additional unlabeled samples could effectively improve the classification performance and robustness. In addition, the training time was slightly longer because of the use of Hessian operators.

As another comparison, the prediction results and errors of ELM, SS-ELM, HELM, and LHSS-ELM are given in Figure 4. The same parameter set was used in the model training, and 500 labeled samples and another 1000 unlabeled samples were used for the entire semi-supervised learning process. LHSS-HELM presents an intuitive improvement in terms of prediction performance.

Furthermore, the comparison of four existing soft sensor methods, namely, semi-supervised probabilistic principal component regression (SS-PPCR),¹⁰ Co-PLS,¹² deep belief network (DBN),¹³ and denoising autoencoder with a neural network (SDAE-NN),¹⁴ were conducted to evaluate the performance of LHSS-HELM. Under the same training conditions, soft sensor models were built. Testing results are presented in Table 3.

Figure 5 shows the comparison results. The SS-PPCR and Co-PLS are semi-supervised approaches

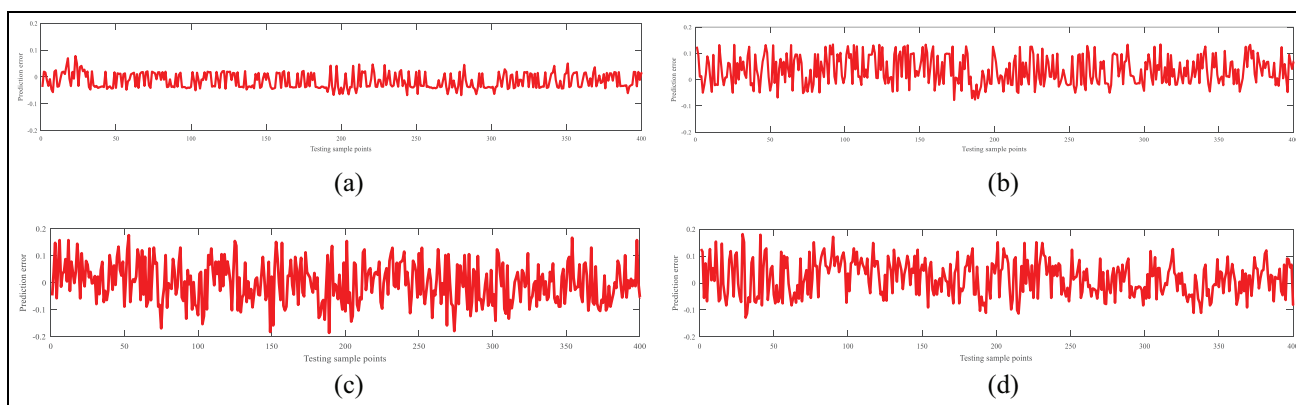


Figure 4. Testing error comparison with the different numbers of unlabeled samples for different algorithms: (a) The proposed method (LHSS-HELM), (b) HELM, (c) ELM, and (d) SS-ELM.

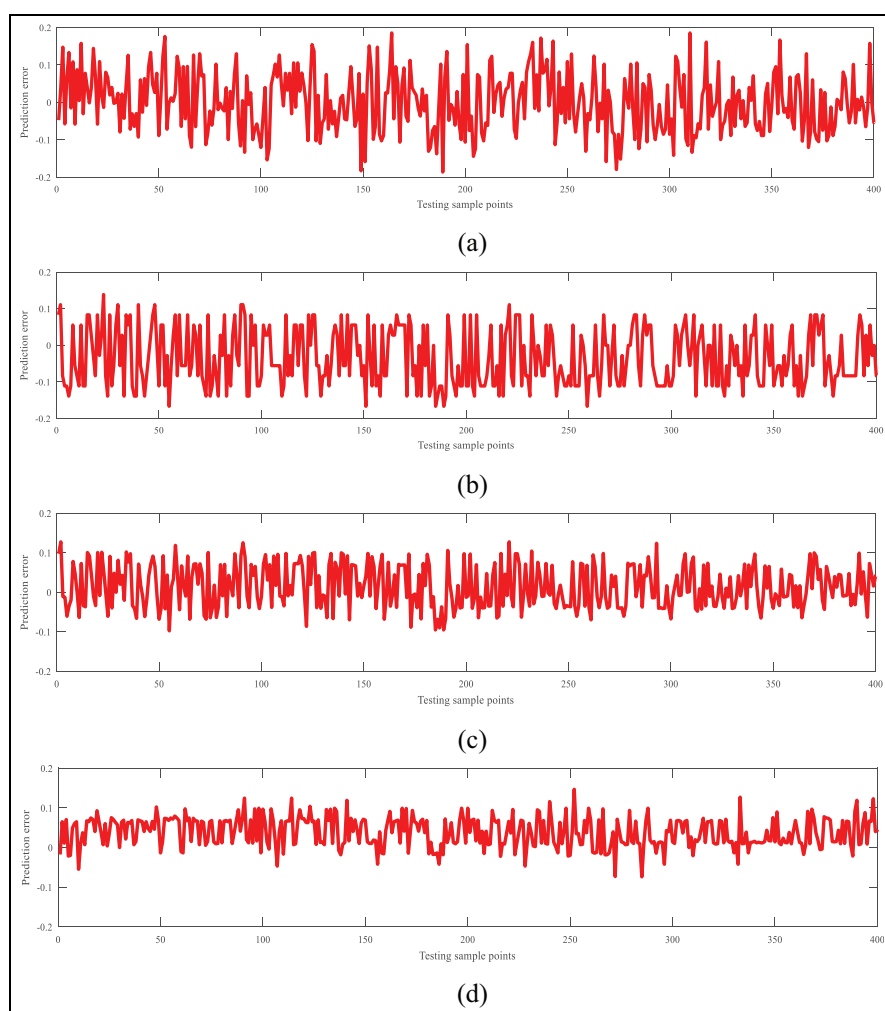


Figure 5. Testing error comparison with different numbers of unlabeled samples for different algorithms: (a) SS-PPCR, (b) Co-PLS, (c) DBN, and (d) SDAE-NN.

that can fully utilize unlabeled data for model training. However, inherent linearity considerably limits prediction performance enhancement. The DBN and SDAE-NN are derived from DL domains, which have been

implemented in some quality variable regression domains. The DBN method is pretrained by a restricted Boltzmann machine²⁵ and then the parameter for initializing BP-NN is obtained. For SDAE-NN, the same

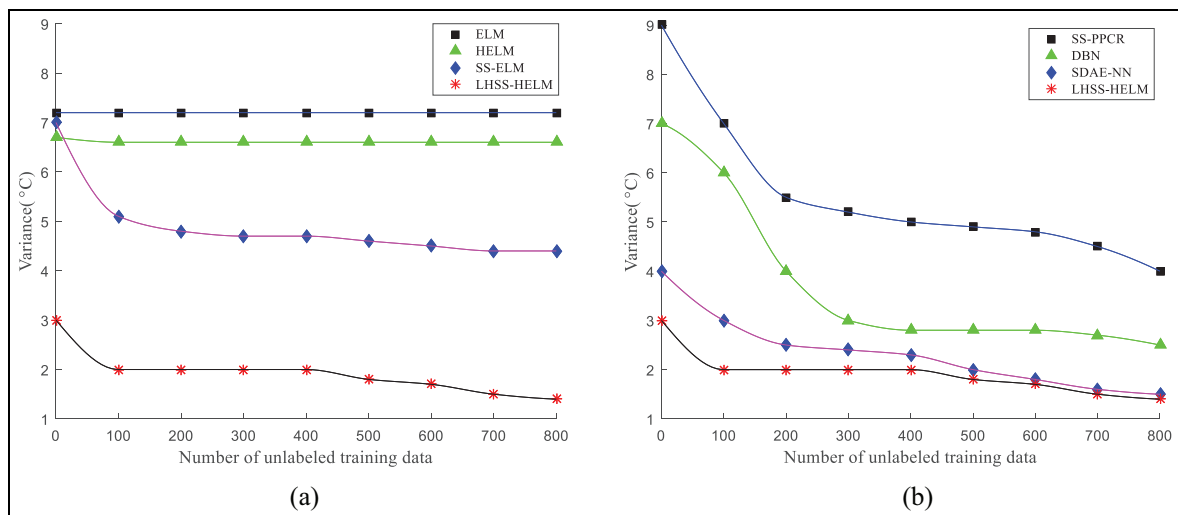


Figure 6. Testing variance comparison with the different number of unlabeled samples for different algorithms: (a) ELM-based comparison and (b) other methods comparison.

method is introduced to obtain the initial weights of NN. For a fair comparison, the number of hidden neurons and hidden layer are the same as with LHSS-HELM. The results demonstrate that both DBN and SDAE-NN have excellent performance for quality prediction, indicating that deep architecture extracts superior and complex features in the high layer. The training speed in DBN and SDAE-NN are slow because the BP algorithm is utilized in model training. All data results in the case showed mean values of 10 iterations of experiments. The variances with different numbers of unlabeled examples are described in Figure 6. The results show that the LHSS-HELM has lower variance and superior robustness to other existing soft sensor methods.

Conclusion

In this study, the DL based on HAE and ELM were utilized to build a semi-supervised soft sensor modeling. Traditional semi-supervised ELM lacks extrapolating power and robustness. To solve this problem, we proposed a weighted Laplacian–Hessian integrated hierarchical SS-ELM algorithm and employed sensor modeling. First, HAEs were used for deep complex feature extraction with all labeled and unlabeled samples. Then, a semi-supervised ELM based on Laplacian–Hessian regularization was used for quality variable regression. To obtain a superior LHSS-HELM model, we proposed a pair of trade-off parameter sequences to update the hyperparameters. After combining multi-layer unsupervised feature extraction and semi-supervised regression, the proposed LHSS-HELM considerably improved the extrapolating ability, performance, and robustness. The soft model for CT prediction implemented in the industrial aluminum reduction cell demonstrated that the performance is superior to the existing state-of-the-art soft sensor

modeling. The extension in LHSS-HELM can be further applied to multiclass classification or regression problems and give new sights for these domains.

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ORCID iDs

Yongxiang Lei  <https://orcid.org/0000-0001-6375-8474>

Hamid Reza Karimi  <https://orcid.org/0000-0001-7629-3266>

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